



Chatbots and the Open Source Tools You Can Use to Develop Them

A chatbot is a computer program or embedded artificial intelligence that can conduct a conversation using auditory or textual methods. Chatbots are also known as smartbots, interactive agents or artificial conversation entities. Let's explore a few of the top open source tools for chatbot development.

The word 'chatbot' is derived from 'chat robot' which allows for highly engaging, conversational experiences, through voice and text. A chatbot can be customised and used on Web browsers, smartphones and other popular messenger-based platforms like Facebook Messenger or Slack. It attempts to simulate the conversation or chatter of a human being through text or voice interactions. A user can ask the chatbot a question or give it a command, and it responds by performing the requested action. Being a computer program, a chatbot is regarded as an example of artificial intelligence. It is typically designed and used in dialogue systems for customer care applications or information-acquisition-cum-knowledge-discovery. Chatbots also make use of natural language processing (NLP) and respond with the most matching

keywords or similar patterns of action from a database.

The concept of the chatbot was proposed by Michael Mauldin in 1994 to define conversational programs. Today, chatbots are popularly accessed through virtual assistants like Google Assistant, Apple Siri, Microsoft Cortana, Amazon Alexa, Samsung Bixby and even Facebook Messenger. The history of the chatbot is quite fascinating—various chatbots have been designed in the past, starting with ELIZA back in 1966, PARRY in 1992, A.L.I.C.E, Jabberwacky and D.U.D.E, etc. Of these, ELIZA and PARRY, which were used for typed conversations, were most popular.

As per the 2017 Global Human Capital Trends Report, 41 per cent of the chief executives of companies claim to have implemented chatbots already or to have prepared cognitive AI

technologies to be implemented in the near future. Another 34 per cent of the companies have already launched the pilot prototypes. According to Gartner, chatbots will be involved in 85 per cent of all types of business-customer interactions by 2020. A report from Radiant Insights states that the chatbot market will grow at a CAGR of 24.43 per cent in the period 2018-2022 due to the increasing popularity of various messaging services. Another report by Transparency Market Research mentions that the global chatbot market in 2015 was valued at US\$ 113 million and the CAGR from 2016 to 2024 will be 27.8 per cent. It has also been predicted that the global chatbot market could reach US\$ 994.5 million by 2024.

The ultimate purpose of a chatbot is to conduct a conversation that allows people to access the desired information with a high level of accuracy using